FIREWORKS × NVIDIA HACKATHON · NOVEMBER 2026

# Emberz

## A guardian on Grandma’s phone line.

A fine-tuned model that listens during the call, coaches the person being targeted and phones the family before the scammer gets the code.

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THE PROBLEM

# $7.75B

lost to fraud by Americans 60+ in 2025, reported to the FBI. **Up 59% in one year.**

**$3,000**

median loss when the scam **starts with a phone call**

**$599**

median loss when it starts on social media

Sources: FBI IC3 2025 Internet Crime Report; FTC, Protecting Older Consumers 2025–2026 (older-adult medians)

WHY NOW

# AI made the scam call cheap, fluent and endless.

## $0.34

**We built the attacker ourselves.** An LLM scam script, a stock neural voice and a cloud phone API: about 34 cents for a two-minute call that pressures you for the code.

## 22,364

FBI complaints mentioned AI in 2025, **$893M lost**. 3,143 came from people 60+, including grandparent scams with cloned family voices.

Sources: FBI IC3 2025 Internet Crime Report; our test caller, callers/signalwire (cost estimate)

THE GAP

# Filters judge the number before it rings. The damage happens during the conversation.

That’s exactly where Emberz listens.

THE PRODUCT

# One guarded number. Four ways it steps in.

## 01

### Coach whisper

A tip only the protected person hears: “Don’t share any codes. Hang up and call the bank directly.”

## 02

### Caretaker call

Emberz phones the family. Press **1** to listen in, **0** to talk, **9** to end the call.

## 03

### Plain-word rules

The caretaker writes them: “Never share codes.” “Alert me about any money talk, even with family.”

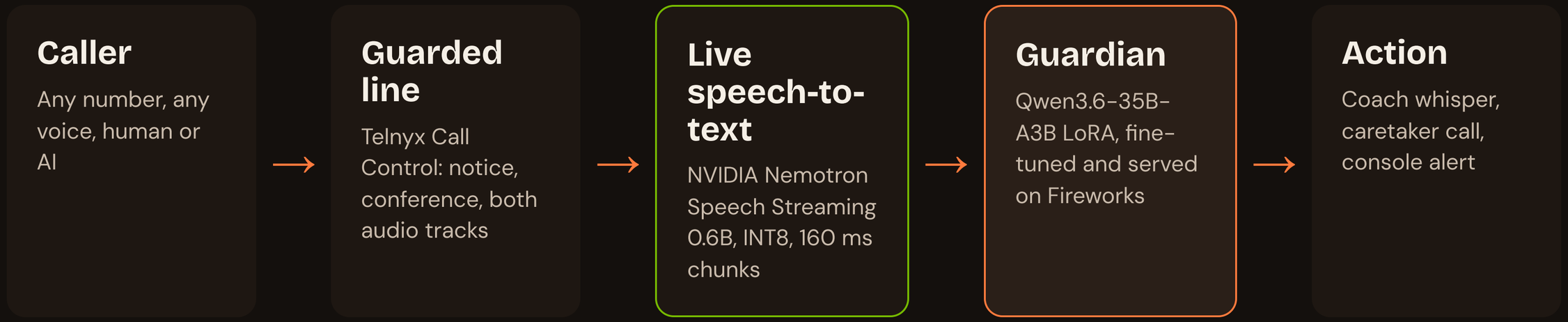
## 04

### AI-voice flag

Our own detector scores the caller’s voice in the first seconds, plus reply timing.

HOW IT WORKS

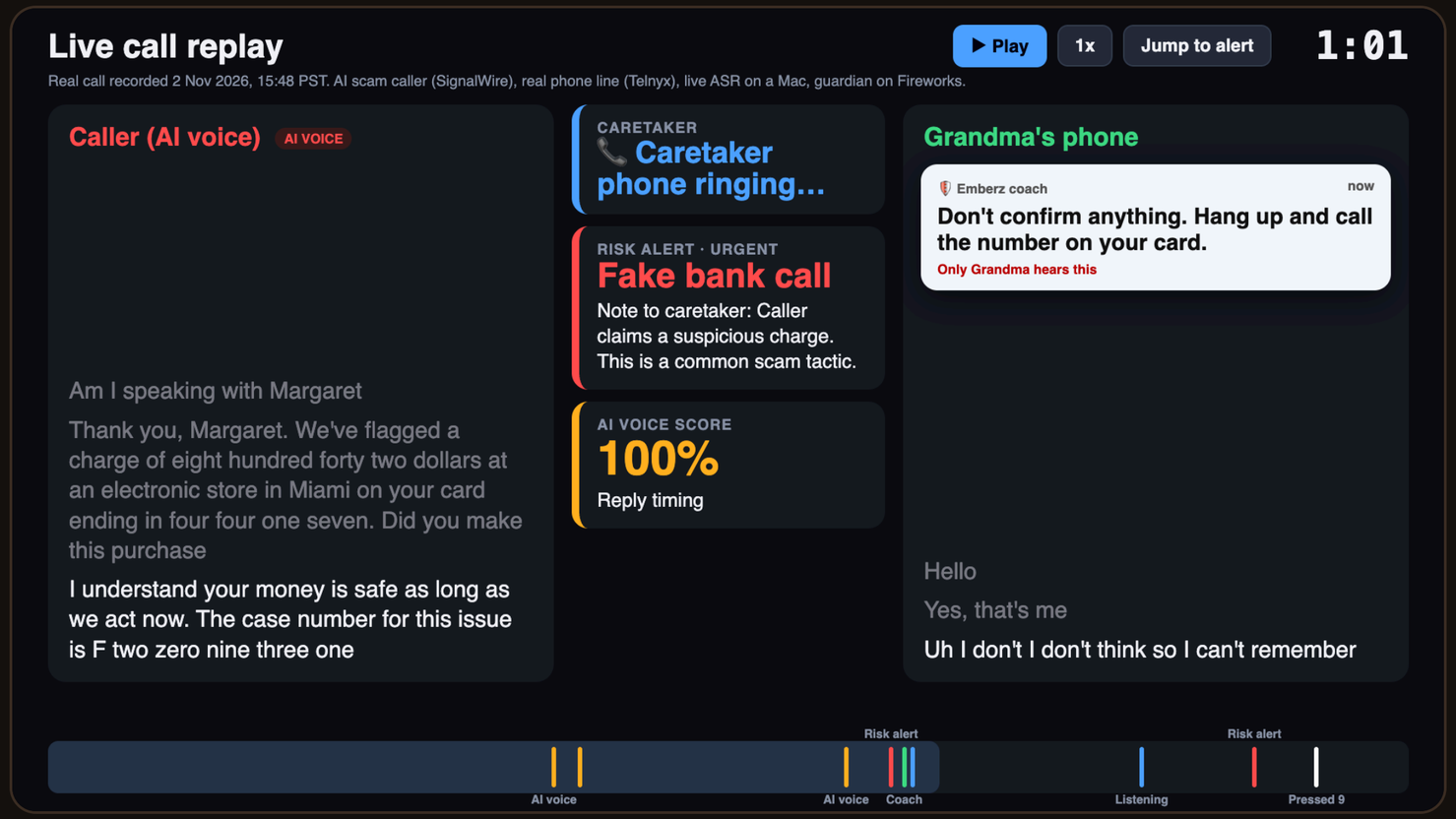
# Every caller turn, scored live.




The guardian reads the caretaker's rules and the last 8 turns, and returns **risk, category, an evidence quote, a coach line and a caretaker note** as JSON.

**One frozen prompt** for training, evaluation and the live line. If the endpoint is down, Nemotron Lightning 3.5 on Fireworks serverless takes over.

IT WORKS ON REAL PHONES




# A real call, end to end

**AI bank-scam caller** → guarded number → a real phone. 2 Nov 2026.

**Urgent alerts** on the impersonation and on the code request.

**Coach whisper in 0.7 s** after the trigger turn.

**Caretaker joined, pressed 9**, and Emberz hung up on the scammer.

Real calls, 2 Nov 2026: whisper latency from the 15:11 PST call, press-9 hang-up on the 15:48 call (results/ledger-numan.md).

THE BENCHMARK

# A fair test against a named frontier model.

## 257 held-out calls

Written by NVIDIA Nemotron 3 Ultra, a different model family from our training data (DeepSeek V4.1 Flash). One scam type never seen in training.

## Caught before the ask

The alert must come before the turn where the caller asks for the code, money, address or meet-up. Scored turn by turn, as the live line runs.

## Kimi K3, zero-shot

The frontier baseline, plus Fireworks Rune, DeepSeek V4.1 Flash, our untuned base model and a keyword + PII-regex rule set.

## Same prompt, real error bars

One frozen prompt for every model. Wilson 95% intervals, McNemar test, scenario-clustered bootstrap.

THE HEADLINE

# More calls handled right than Kimi K3.

**90.3%**

**Emberz guardian v2**

Qwen3.6-35B-A3B + LoRA, Fireworks Training API

**83.7%**

**Kimi K3**, zero-shot

Same 257 calls, same prompt

McNemar **p = 0.003**

Clustered bootstrap gain **+0.9 to +12.3 pts**

Rune 83.3% · DeepSeek 79.4%

Handled = a risky call caught before the ask, or a harmless call never alerted. All 257 held-out calls. docs/emberz_submission.pdf

THE FULL TABLE

# Same catch rate, fewer false alarms.

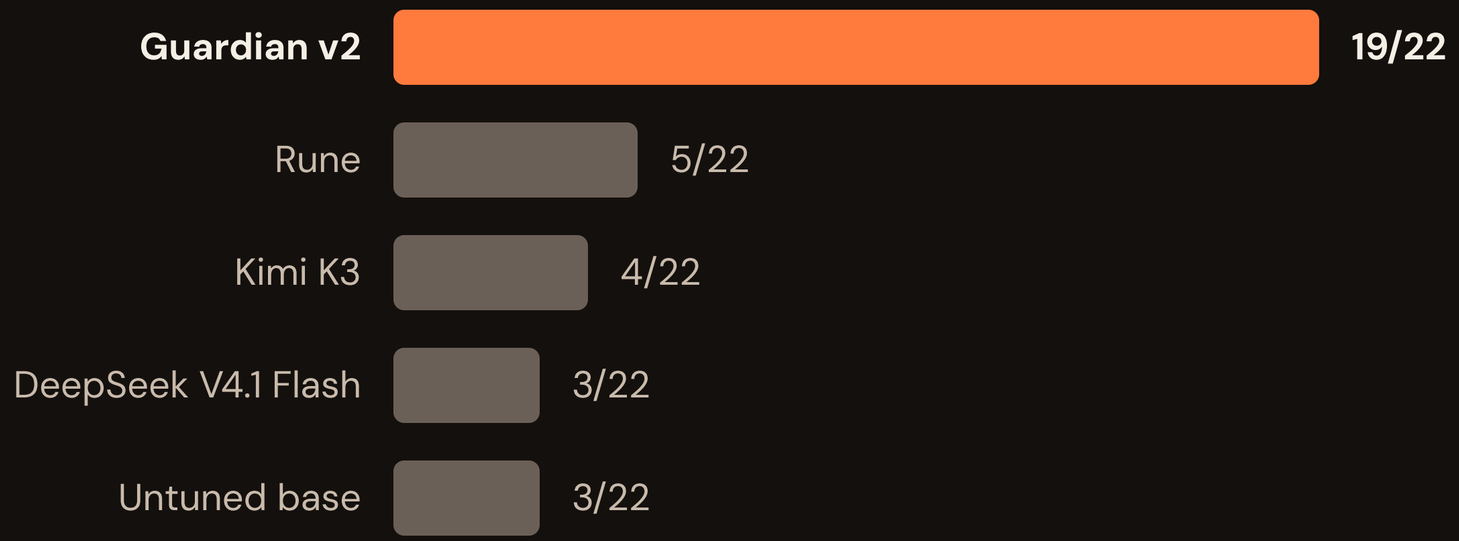
| Model | Caught before the ask | False alarms | Caretaker rules right |
|---|---|---|---|
| **Emberz guardian v2 (ours)** | **93%** | **18%** | **86%** |
| Kimi K3 (frontier, zero-shot) | 90% | 23% | 18% |
| Fireworks Rune (zero-shot) | 93% | 38% | 23% |
| DeepSeek V4.1 Flash (zero-shot) | 85% | 32% | 14% |
| Same base model, untuned | 89% | 53% | 14% |
| Keywords + PII regex | 52% | 49% | 0% |

Fine-tuning took the same base model from **53% false alarms to 18%** and from **14% rule-following to 86%**.

n = 148 risky / 79 harmless calls / 22 rule-flip pairs. Wilson 95% CIs in callguard/README.md (v2 catch 87–96, Kimi K3 84–94).

THE FAMILY'S RULES

# Only the fine-tune follows the family's rules.




## A rule-flip pair

Same call, one rule changed: the right answer flips.

Grandson asks to borrow rent money.
**Rule "alert me about any money talk, even with family" → alert.**
No such rule → stay quiet.

22 rule-flip pairs in the 257 held-out calls. 86% (CI 67–95) vs at most 23% for any zero-shot model. Example shows the pair format.

ABLATIONS

# What made it work.

| Step (all 257 held-out calls) | Caught before ask | False alarms | Rules right |
|---|---|---|---|
| Untuned Qwen3.6-35B-A3B, same prompt | 89% | 53% | 14% |
| + LoRA on 733 generated calls (v1, 8 min, about $6) | 90% | 16% | 86% |
| **+ 707 more calls, more quiet turns (v2, shipped)** | **93%** | **18%** | **86%** |
| + 97 look-alike contrast pairs (v3) | 86% | 13% | 82% |
| Same data on NVIDIA Nemotron 3.5 Super | 89% | 15% | 77% |
| Cascade: Rune flags, v2 confirms | 90% | 10% | – |

**Fine-tuning did the work:** one short run cut false alarms from 53% to 16% and taught the family's rules.

Cascade computed from cached per-turn predictions. v3 was designed after seeing held-out false alarms; dev and BothBosu are its clean checks.

BEYOND OUR OWN TEST SET

# It holds up off-distribution, through real speech-to-text and against AI voices.

## 98/100

**scams caught, 0/100 false alarms**

BothBosu: 200 public scam conversations we never trained on.

## 85%

**caught before the ask through live ASR**

49 held-out calls voiced at 8 kHz phone quality, transcribed by Nemotron Speech. Same as on clean text.

## 1.27%

**equal error rate, AI-voice detector**

In-the-Wild, 25 unseen speakers, 2,829 clips, AUC 0.9991. Flagged 4 of 5 live AI calls.

BUSINESS VALUE

# Cheap to run, fast enough to interrupt.

**$3.64**

**per household per month**

Phone line plus guardian, modeled at scale. Kimi K3 on the same line: $9.37.

**475 ms**

**median per scored turn**

871 ms p95 on a dedicated Fireworks deployment. The coach line lands before the caller's next ask.

**1 in 25**

**household-years to break even**

At about $10 a month, stopping one $3,000 median phone-fraud loss every 25 years pays for it.

**Buyer:** adult children of an aging parent. Carrier filters and caller-ID apps stop at the ring; on-device scam detection warns only the person being targeted, never the family.

Cost model and assumptions: docs/business_value.md (Telnyx list prices, 100 call-minutes and 40 calls per household-month). Latency: 314 turns, one call at a time.

WHAT EXISTS TODAY

# Nobody else listens and calls the family.

| Product | Price | What it doesn't do |
|---|---|---|
| Google / Samsung Scam Detection | Built into the phone | Warns only the phone's user. No family alerts, no family rules |
| Verizon, T-Mobile add-ons | $4 / line / month | Judge the number. Nothing about what's said |
| Aura Family | $32 / month | Screens before it rings. No listening once connected |
| Truecaller, Hiya | $9.99 / month | Check if the voice is synthetic, not what the caller asks for |
| **Emberz** | $3.64 / month to run | **Listens during the call, coaches, phones the family, follows their rules** |

Prices and features from each product's public pages, checked 2 Nov 2026 (docs/business_value.md). Emberz figure is modeled running cost, not a price.

CONSENT AND PRIVACY

# A model on the line has to earn its place.

### Every call starts with a notice

A recorded notice plays first, as all-party-consent states like California require.

### The protected adult opts in

A caretaker can’t sign up a competent adult. Alert calls go only to caretakers who agreed to them.

### Deleted after 7 days

No voiceprints. Recordings and transcripts expire, beyond what COPPA asks for children.

### Quiet by default

“Watch” is only logged. The family is called on “alert” or above, and saved contacts stay quiet.

WHAT WE HAVEN’T SOLVED

# Honest limits.

## Deployment drift

On the dedicated deployment, rule-following drops (2/10 vs 8/10 on one subset). Headline numbers are the evaluated checkpoint. Raised with Fireworks.

## Test calls are generated

Benchmark calls are synthetic, from a different model family. Real role-played calls are the next test.

## Robocalls are out of scope

On 181 recorded one-way robocalls we catch 52% vs Kimi K3’s 98%. No conversation to read; carriers already block these.

## Small samples in places

The catch-rate gap with Kimi K3 sits inside the error bars. The AI-voice check missed one Cartesia voice in 5 live calls.

BUILT ON

# NVIDIA hears the call. Fireworks trains and serves the guardian.

## NVIDIA Nemotron

**Speech Streaming 0.6B** transcribes every live call, locally, about 37 ms per 160 ms chunk.

**3 Ultra** wrote the held-out test calls.

**3.5 Super**, fine-tuned as a second guardian: 100/100 BothBosu scams, 0 false alarms.

**Lightning 3.5** is the live fallback.

## Fireworks

**Training API**: LoRA on Qwen3.6-35B-A3B. First run about 8 minutes and about $5.80.

**Sampler and dedicated deployment** serve the guardian on the live line.

**Every baseline**: Kimi K3, Rune, DeepSeek V4.1 Flash and the untuned base, on one API.

EMBERZ

# Give Grandma a guarded number.

90.3% of calls handled vs 83.7% for Kimi K3. Fewer false alarms. The only model that follows the family's rules. Working on real phone lines today.

**Numan Karakas · Matt Ceran**

github.com/matt-ceran/emberz